Downloaded from SAE International by American Univ of Beirut, Sunday, July 29, 2018

SAE TECHNICAL PAPER SERIES

2000-01-0815

A New EPS Control Strategy to Improve Steering Wheel Returnability

Masahiko Kurishige, Shunichi Wada, Takayuki Kifuku, Noriyuki Inoue, Ryoji Nishiyama and Shigeki Otagaki
Mitsubishi Electric Corp.

Reprinted From: Steering and Suspension Technology Symposium 2000 (SP–1519)

SAE The Engineering Society For Advancing Mobility Land Sea Air and Space® INTERNATIONAL

SAE 2000 World Congress
Detroit, Michigan
March 6–9, 2000

400 Commonwealth Drive, Warrendale, PA 15096-0001 U.S.A. Tel: (724) 776-4841 Fax: (724) 776-5760

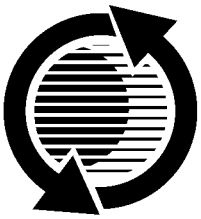
Downloaded from SAE International by American Univ of Beirut, Sunday, July 29, 2018

The appearance of this ISSN code at the bottom of this page indicates SAE's consent that copies of the paper may be made for personal or internal use of specific clients. This consent is given on the condition, however, that the copier pay a $7.00 per article copy fee through the Copyright Clearance Center, Inc. Operations Center, 222 Rosewood Drive, Danvers, MA 01923 for copying beyond that permitted by Sections 107 or 108 of the U.S. Copyright Law. This consent does not extend to other kinds of copying such as copying for general distribution, for advertising or promotional purposes, for creating new collective works, or for resale.

SAE routinely stocks printed papers for a period of three years following date of publication. Direct your orders to SAE Customer Sales and Satisfaction Department.

Quantity reprint rates can be obtained from the Customer Sales and Satisfaction Department.

To request permission to reprint a technical paper or permission to use copyrighted SAE publications in other works, contact the SAE Publications Group.



GLOBAL MOBILITY DATABASE

All SAE papers, standards, and selected books are abstracted and indexed in the Global Mobility Database

No part of this publication may be reproduced in any form, in an electronic retrieval system or otherwise, without the prior written permission of the publisher.

ISSN 0148-7191
Copyright © 2000 Society of Automotive Engineers, Inc.

Positions and opinions advanced in this paper are those of the author(s) and not necessarily those of SAE. The author is solely responsible for the content of the paper. A process is available by which discussions will be printed with the paper if it is published in SAE Transactions. For permission to publish this paper in full or in part, contact the SAE Publications Group.

Persons wishing to submit papers to be considered for presentation or publication through SAE should send the manuscript or a 300 word abstract of a proposed manuscript to: Secretary, Engineering Meetings Board, SAE.

Printed in USA

Downloaded from SAE International by American Univ of Beirut, Sunday, July 29, 2018

2000-01-0815

A New EPS Control Strategy to Improve Steering Wheel Returnability

Masahiko Kurishige, Shunichi Wada, Takayuki Kifuku, Noriyuki Inoue, Ryoji Nishiyama and Shigeki Otagaki
Mitsubishi Electric Corp.

Copyright © 2000 Society of Automotive Engineers, Inc.

ABSTRACT

This paper proposes a new Electric Power Steering (EPS) control strategy that enables improvement to steering-wheel returnability. Using a conventional EPS controller, frictional loss torque in the steering mechanism reduces steering-wheel returnability, which drivers occasionally perceive as unpleasant. This phenomena occurs in any EPS system regardless of motor type or mounting location. To improve steering-wheel returnability for EPS-equipped vehicles, we developed a new control strategy based on estimation of alignment torque generated by tires and road surfaces. This proposed control strategy requires no supplemental sensors like steering-wheel angle or motor-angle sensors. We experimented with this proposed control algorithm using a test vehicle and confirmed that it enables improved steering wheel returnability and also better on-center feeling.

INTRODUCTION

Recently, vehicles equipped with power steering have come into widespread use, and nowadays even midget passenger cars provide them. Most power steering systems are hydraulically power assisted. However, the fuel consumption of vehicles equipped with a hydraulic power-steering system is worse than that of manual steering systems because oil pressure must be continuously boosted. EPS systems are on-demand systems that operate only when the driver requires. Hence, using EPS enables us to improve fuel consumption and reduce emissions. This has encouraged the present great increase in EPS equipped vehicles.

Power-steering systems reduce the driver's steering effort for parking-lot maneuvers and provide a solid steering feeling to drivers during freeway and street driving. In these ways, power-steering systems mainly work when steering torque is applied. In addition to these driving conditions, in which steering torque is applied to the steering wheel, power-steering systems must also meet the requirement for driving conditions when no torque is applied to the steering wheel. One of these requirements is vehicle stability when the driver releases the steering wheel just after normal steering during high-speed driving, and another is steering-wheel returnability when the driver releases the steering wheel immediately after a slight steering adjustment during high-speed driving or just after normal steering during relatively low-speed driving. Using a conventional EPS controller, frictional loss torque reduces steering-wheel returnability, and drivers occasionally become aware of this effect.

To overcome this effect, we have developed a new control strategy to improve returnability of the steering wheel using a newly developed means of observing the alignment torque from the tires. This control strategy provides good estimates of the alignment torque even if the lateral acceleration of the vehicle is small. Dependent on the estimated alignment torque, the current of the motor is compensated as if the alignment torque were increased under the limiting condition of small lateral vehicle acceleration.

We have tested this novel control strategy and confirmed its effectiveness in improving steering-wheel returnability. Using this control strategy, increasing the torque gradient around 0 G also caused a remarkable improvement in "on-center" feeling. We have tested the effects under various driving conditions and confirmed that no harmful subjective effects were observed.

This proposed control strategy requires no supplemental sensors such as steering-wheel angle or motor-angle sensors.

GOAL OF NEWLY DEVELOPED CONTROLLER

ON-CENTER FEELING - One significant factor in evaluating steering performance is on-center handling, which indicates the sensation of lane-change maneuvers and that of steering adjustments necessary to follow the lane accurately. To ease drivers' on-center handling maneuvers, "steering-torque gradient" and "steering-

Downloaded from SAE International by American Univ of Beirut, Sunday, July 29, 2018

wheel returnability".[1] play important roles. The steering torque gradient is the differential of steering torque with respect to vehicle lateral acceleration. The steering-wheel returnability indicates the lateral acceleration (m/s^2) when the steering torque is around 0 (Nm). Larger steering-torque gradients and smaller steering-wheel returnability definitely provide better steering sensations for drivers.

TARGET OF STEERING PERFORMANCE - The goal of this study is to develop a controller that clearly provides improved steering-torque gradient and steering-wheel returnability without using supplemental sensors such as steering-wheel angle sensors or motor-angle sensors. Also, freedom from harmful subjective effects is essential, that is to say, reliability is the most important factor.

STEERING MECHANISM WITH EPS

STEERING MECHANISM - In an EPS system, a motor is applied via a reduction gear to the steering mechanism. There exist both frictional loss torques, and spring characteristics or backlash in the steering mechanism[2]. When a driver steers, alignment torque is generated by tires and road surfaces. When a vehicle is running at a constant speed, the alignment torque is nearly proportional to the vehicle lateral acceleration under on-center handling conditions. The reason for the proportional relationship between them is as follows. The alignment torque is the product of side force and total length of caster trail and pneumatic trail. Here, pneumatic trail varies depending on the slip angle of the tire while caster trail is constant. For on-center handling condition, the pneumatic trail is practically constant. Hence, the alignment torque is nearly proportional to the slip angle as shown in figure1a although the linear region of alignment torque is smaller than the linear region of side force as shown in figure1b because pneumatic trail begins to drop beyond the linear region. Further, the vehicle lateral acceleration is roughly proportional to the side force of the front wheels for on-center handling condition. Hence, the alignment torque is nearly proportional to the vehicle lateral acceleration and associated steering wheel angle for on-center handling conditions. However, if the vehicle lateral acceleration is large or the road is slippery, the alignment torque will saturate and eventually decrease with further increases in vehicle lateral acceleration. The proposed EPS controller uses the newly developed alignment torque observer which estimates the alignment torque for the on-center regions. In this region, the alignment torque is relatively small as shown in figure1a. Hence to estimate it precisely, an accurate steering model is required. This will be discussed briefly below.

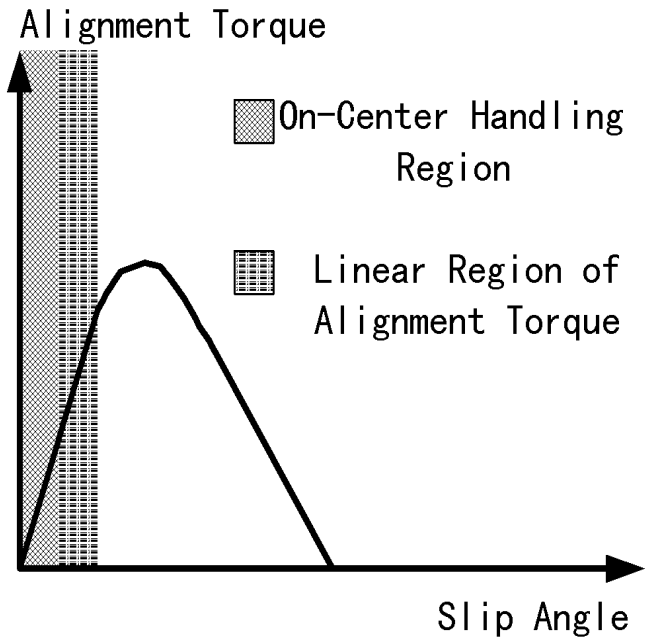


Figure 1a. Slip Angle vs. Alignment Torque

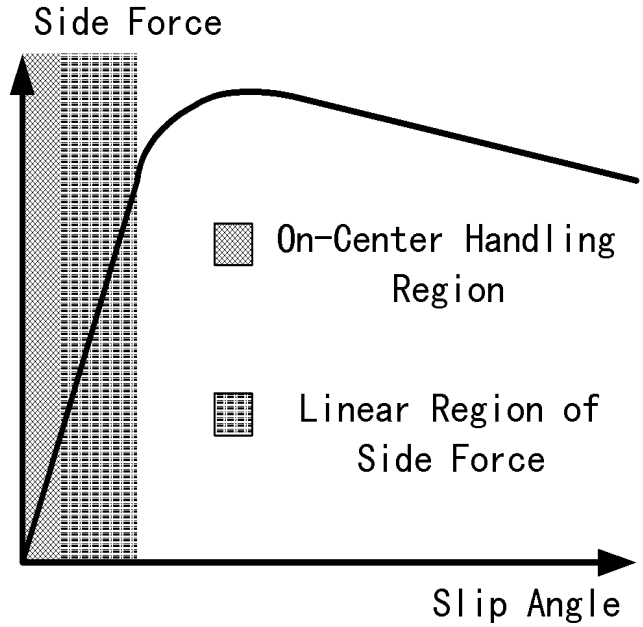


Figure 1b. Slip Angle vs. Side Force

Next, EPS operation is explained referring the Figure2. When a driver steers, the steering torque is detected by a torque sensor that is placed between the steering wheel and motor. According to the torque sensor signal, motor current is determined. The output torque of the motor is input to the steering shaft through a reduction gear and the product of motor torque and gear ratio act as torque assistance for the driver. The sum of the driver's steering torque and this assisting torque rotates the steering shaft against the alignment torque from the tires and the frictional losses in the steering mechanism. To control the EPS system, the supply voltage and current to the motor are also detected in addition to the steering torque.

Downloaded from SAE International by American Univ of Beirut, Sunday, July 29, 2018

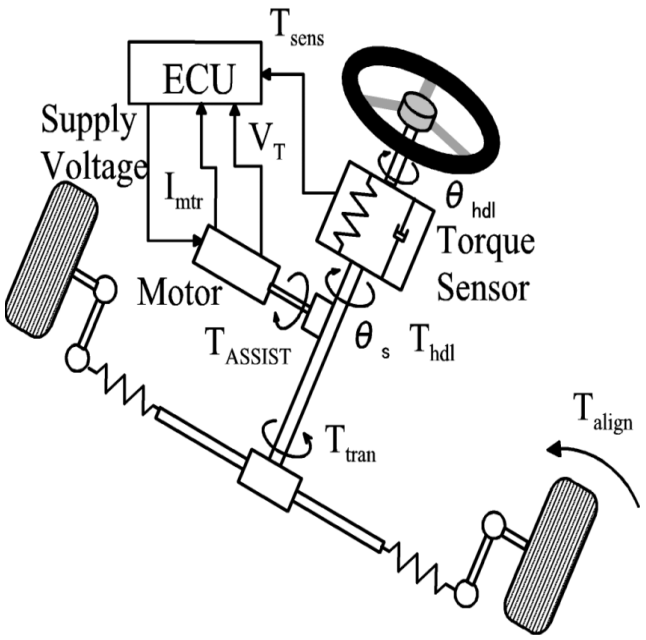


Figure 2. Steering Mechanism with EPS

MODEL OF STEERING MECHANISM - The steering mechanism is expressed as Equation 1[3]. "T_{tran}" denotes reaction torque to the steering shaft and this is the sum of alignment torque from tires "T_{align}" and frictional loss torque of the rack & pinion etc "T_{f_rp}".

$$J\ddot{\theta}_s = T_{ASSIST} + T_{hdl} - T_{tran} \tag{1a}$$

where

$$T_{ASSIST} = G_{gear} \cdot \{K_T \cdot I_{mtr} - T_{f_mtr} \cdot \mathrm{sgn}(\dot{\theta}_s)\}$$
$$T_{tran} = T_{align} + T_{f_rp} \cdot \mathrm{sgn}(\dot{\theta}_s) \tag{1b}$$

Equations 1a and 1b yield the equation 2.

$$J\ddot{\theta}_s = T_{hdl} + G_{gear} \cdot K_T \cdot I_{mtr} - T_{align} - (G_{gear}^2 \cdot T_{f_mtr} + T_{f_rp}) \cdot \mathrm{sgn}(\dot{\theta}_s) \tag{2}$$

When no steering torque is input by the driver, the behavior changes depending on the alignment torque "T_{align}" and total frictional losses" $G_{gear}^2 T_{f_mtr} + T_{f_rp}$". In cases where the alignment torque is smaller than the total frictional loss torque, the steering wheel will not return and the vehicle may move in an unexpected direction until the driver himself forces its return by the steering wheel. This phenomenon often occurs especially just after normal steering during low speed driving or steering adjustments made to follow the lane during high speed driving. Here, T_{hdl} is the steering torque and I_{mtr} is the current of the motor. Both of them are measurable in EPS systems. The angular-acceleration of the motor in the left-hand term can be estimated precisely for drivers' steering frequency without using an angular-velocity sensor.

CONTROL STRATEGY

ALIGNMENT TORQUE OBSERVER. The estimation algorithm – The newly developed alignment torque observer estimates the alignment torque based on the model of the steering mechanism shown in equation 2. The non-linear term is dealt with by applying the describing function method[4].

The estimation result – An example of estimation result of the alignment torque is shown in figure 3. In this example, the driver steered a sinusoidal pattern on the assumption that the vehicle is changing lanes. The peak lateral acceleration is from 0.1g to 0.15g. The upper row denotes the measured alignment torque while the lower row denotes the estimated alignment torque. The measured alignment torque was obtained by multiplying the lateral acceleration and the gain which convert the lateral acceleration to the alignment torque. This conversion is effective when vehicle lateral acceleration is low and the driver is steering slowly.

As will be clear from figure 3, the alignment torque observer estimates the alignment torque well.

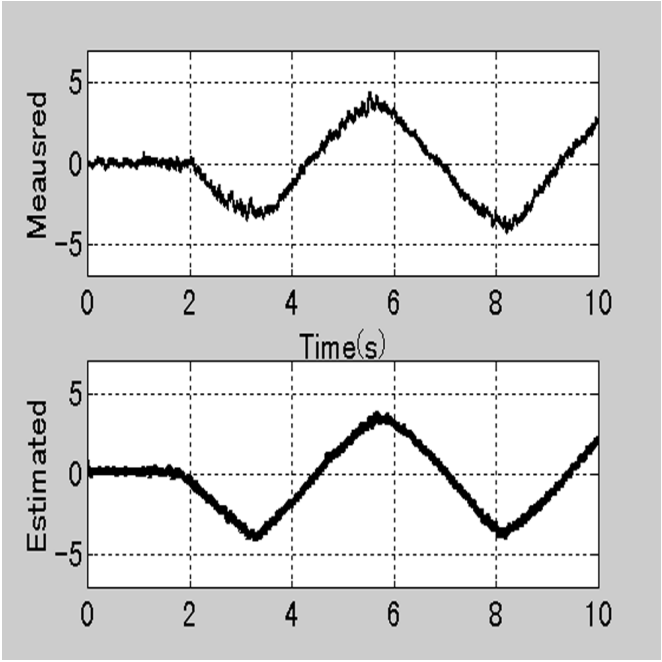


Figure 3. The Estimation Result for Alignment Torque

CONTROLLER – The proposed controller works to augment the returning torque of the steering wheel when the alignment torque is small. This controller works based on the estimated alignment torque using the above alignment torque observer. The block diagram of this controller is shown in figure 4. The controller was designed to augment the returning torque of the steering wheel so as to exceed the total frictional loss torque in the steering mechanism. This augmented returning torque remarkably improves steering-wheel returnability for on-center handling conditions.

Downloaded from SAE International by American Univ of Beirut, Sunday, July 29, 2018

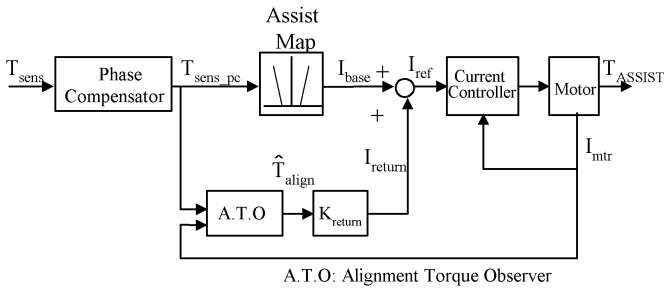


Figure 4. The block diagram of the proposed controller

EXPERIMENT

BENCH TEST – At first, we bench tested the performance of this controller. The characteristic of this bench was tuned to be equivalent to that of vehicle running at about 30km/h. In this experiment, the steering pattern is release of the steering wheel just after a slight steering adjustment.

The results of the experiments are shown in figure 5. In each figure, the upper row denotes the steering wheel angle and the lower row denotes the steering torque. Figure 5a shows the result with the conventional controller. The steering wheel was stuck and would not return to the center. Figure 5b denotes the result with the proposed controller. The steering wheel returns to the center immediately when the driver releases it. Figure 5c denotes the result with the proposed controller, which was tuned to provide excessive returning torque. The steering wheel returns over the center position. From these bench-test results, the effectiveness of this proposed controller to improve returnability and the tunability of the degree of return were both confirmed. These results encouraged us to proceed to experiments using a test vehicle.

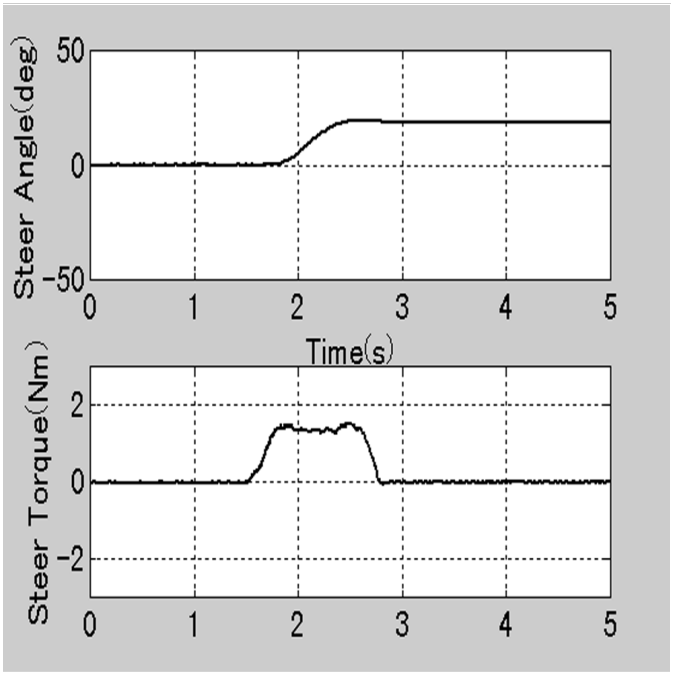


Figure 5a. Bench test result with conventional controller

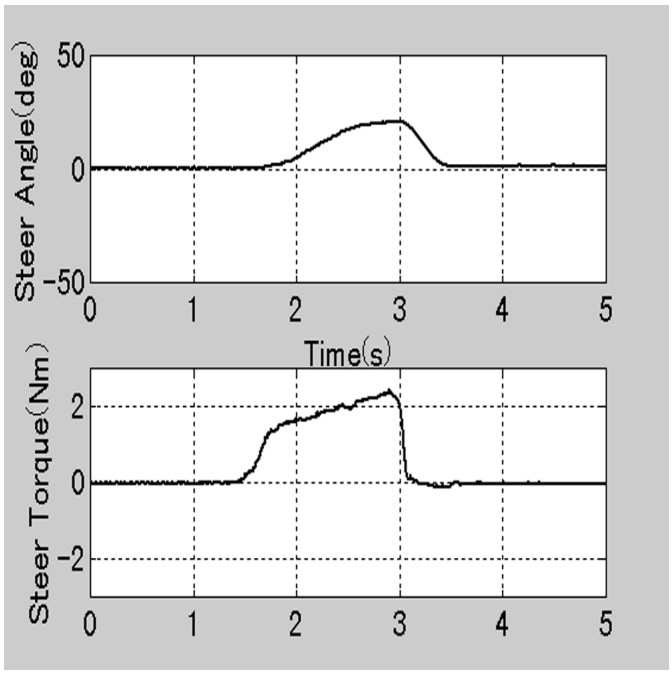


Figure 5b. Bench test result with proposed controller

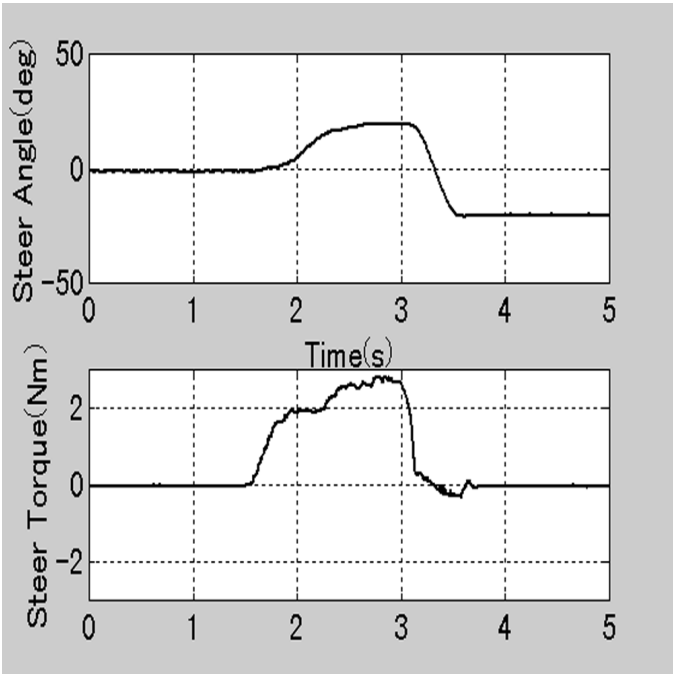


Figure 5c. Bench test result with proposed controller tuned to provide excessive returning torque

VEHICLE TEST. Test condition – The test vehicle is a compact car with a 1000cc SI engine. In this experiment, the steering patterns we selected were release of the steering wheel just after a slight steering adjustment and sinusoidal steering. Both of them were performed at 40km/h.

Test results - An typical test result for sinusoidal steering is shown in figure6a. In this example, the driver steered on the assumption that the vehicle was changing lanes. The peak lateral acceleration was about 0.15g. The x-axis denotes the lateral acceleration and y-axis denotes

Downloaded from SAE International by American Univ of Beirut, Sunday, July 29, 2018

the steering torque. The gradient around 0G indicates the road feel and the larger gradient definitely provides better steering sensations for drivers.

Figure6a shows the result with the conventional controller and figure6b shows the result with the proposed controller. Using the proposed controller, the torque gradient increases, which provides improved steering feeling. Also, using the proposed controller, the transverse width of the hysteresis curve is reduced about the x-axis i.e. the steering torque is 0Nm. The reduction of this width definitely provides improved steering-wheel returnability. To confirm the improved steering-wheel returnability, a typical test result for the release of the steering wheel at 0(sec) is shown in figure7. Using the proposed controller, the residual steering wheel angle at 0.4(sec) is smaller than that using the conventional controller. As mentioned in the “BENCH TEST” section, the residual steering-wheel angle can be readily adjusted by tuning the controller parameters. In this test vehicle, the residual angle was adjusted to provide the best feeling for drivers.

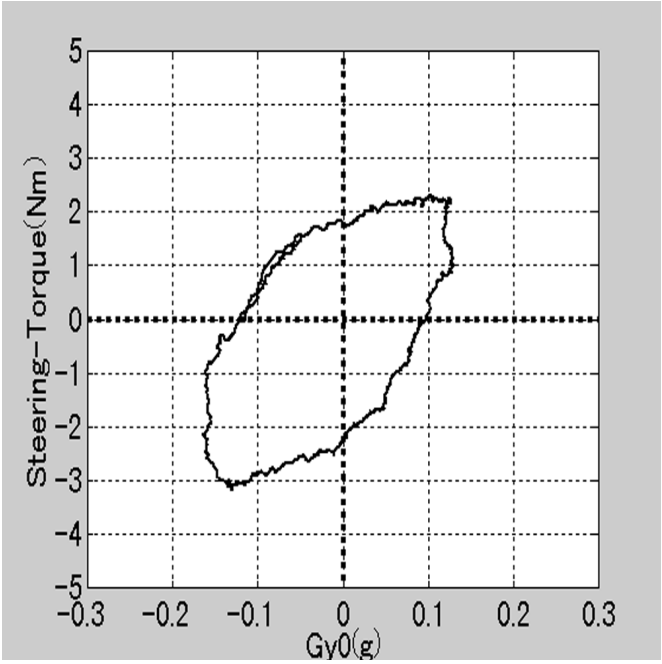


Figure 6a. Vehicle test of sinusoidal steering with the conventional controller

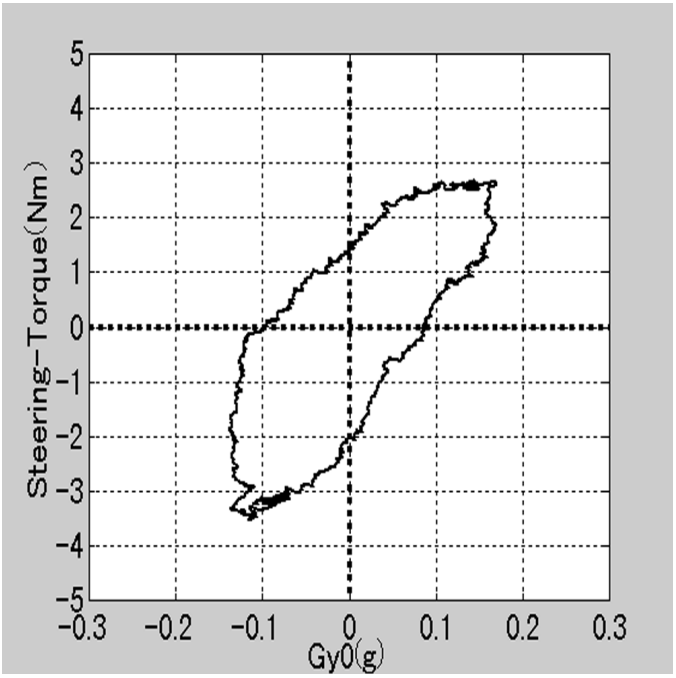


Figure 6b. Vehicle test of sinusoidal steering with the proposed controller

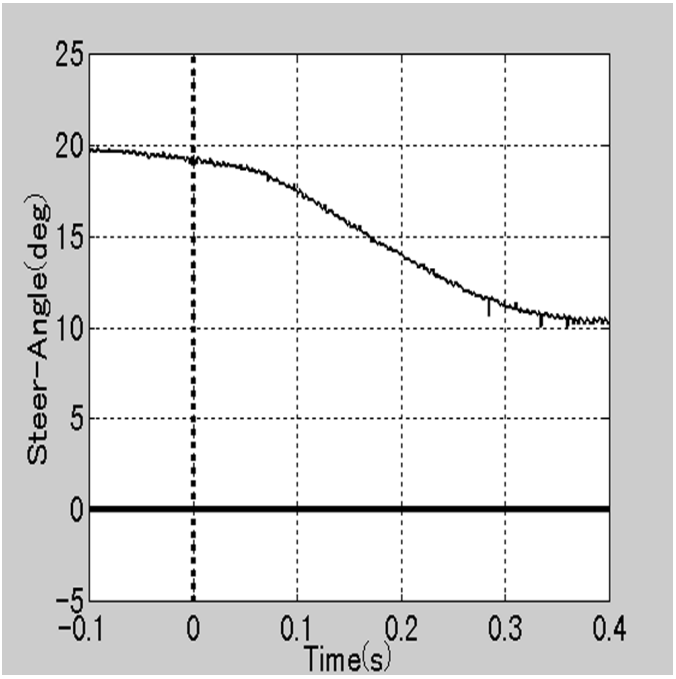


Figure 7a. Vehicle test of releasing the steering wheel

Downloaded from SAE International by American Univ of Beirut, Sunday, July 29, 2018

with the conventional controller

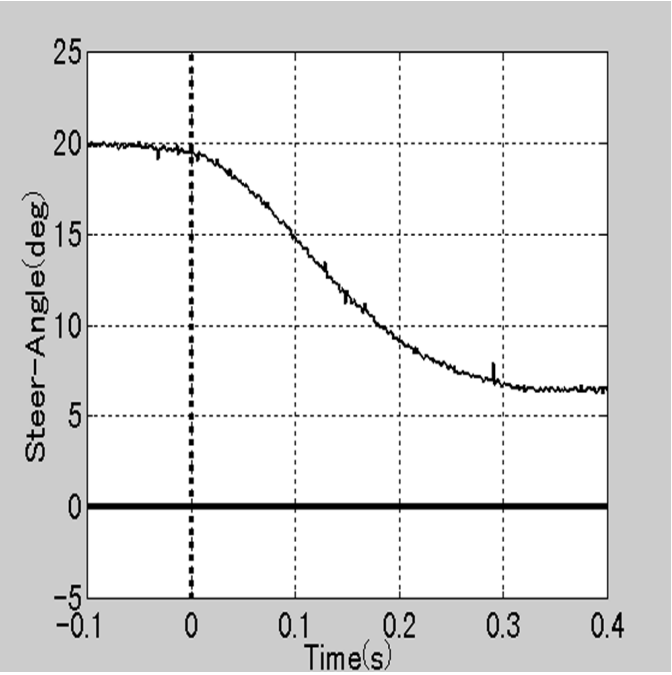


Figure 7b. Vehicle test of releasing the steering wheel with the proposed controller

Further test results – To confirm the absence of inherently harmful subjective effects, we tested under various driving conditions including following a road with deep ruts, a dirt road, freeway and street using various steering patterns. The drivers perceived no unnatural feelings.

NOMENCLATURE

T_{sens} :Measured steering torque

I_{mtr} :Motor current

V_t :Terminal voltage of Motor

q_{hdl} :Steering wheel angle

q_s :Column axis angle

J :Moment of Inertia of motor (at column axis)

T_{hdl} :Driver's steering torque

T_{tran} : Amount of reaction torque from downstream of the column (rack & pinion , tire ,etc)

K_T :Torque constant of the motor

T_{f_mtr} :Frictional loss of the motor

T_{f_rp} :Frictional loss of the rack and pinion

G_{gear} :Reduction gear ratio from motor to column axis

T_{ASSIST} :Assist torque by motor (at column axis)

T_{align} :Reaction torque of tire (at column axis, with ^ : estimated)

T_{sens_pc}:Measured steering torque(phase compensated)

K_{return} :The gain of auxiliary controller

I_{ref} :Motor reference current

I_{base} :Indicated current to assist drivers' steering torque based on the assist map

I_{return} :Indicated current of the auxiliary controller to return the steering wheel

CONCLUSION

The new control strategy to improve steering-wheel returnability was studied with the following results.

1. The proposed control algorithm improves steering-wheel returnability remarkably.
2. The proposed controller also improves on-center feeling.
3. The proposed controller requires no supplemental sensors like steering wheel angle sensors or motor angle sensors.
4. Using the proposed controller, no harmful subjective effects occur.

ACKNOWLEDGMENTS

The success of developing this new controller is attributed to the assistance from members of Himeji Works, Mitsubishi Electric Corp. We would like to thank them.

REFERENCE

1. Norman K.D. et. al., "Objective Evaluation of On-Center Handling Performance", SAE Paper 840069, 1984
2. Post J. W., "Modeling and Testing of Automobile Power Steering Systems for the Evaluation of On-Center Handling", Ph.D. Dissertation, Clemson University, 1995.
3. Kurishige M. et. al., "A Control Strategy to Reduce Steering Torque for Stationary Vehicles Equipped with EPS", SAE Paper 1999-01-0403, 1999
4. Gibson J.E., "Nonlinear Automatic Control, McGraw-Hill, 1963